

Science

Chemistry Lessons
Chemistry Labs
Nature Notebook
Nature Walks & Scouting





About the Course

This course includes the following topic(s): Chemistry Lessons, Chemistry Labs, Nature Notebook: Grades 9-12, Nature Walks & Scouting: Grades 9-12

About Chemistry Lessons

Alveary High School Chemistry provides a natural next step following Physical Science (e.g., Alveary Form 3 Science) and establishes a solid foundation for High School Biology. The course progresses through all expected topics in introductory chemistry and incorporates living engagement and special attention to citizenship, problem-solving, and communication skills for a complete Charlotte Mason science course. This course includes our video companion series, currently at no additional cost.

About Chemistry Labs

Labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method. Labs for this course are integrated into the lessons to facilitate adequate time for more involved activities and to better coordinate with the lessons.

About Nature Notebook: Grades 9-12

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

About Nature Walks & Scouting: Grades 9-12

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.



Placement & Combining Tips

Chemistry Lessons

Recommended for learners who have completed Physical Science, such as Alveary Grade 7-8 Science, including the laboratory activities. Learners should be taking at least Algebra 1 alongside Chemistry (Algebra 2 if using Denison Success). Teachers wishing to place students in Physical Science instead of Chemistry may choose either to complete Alveary Grade 7-8 Science in a single year (with 5 lessons + 1 lab each week available in the Grade 7-8 Quick Links) or to purchase separately from Classical Academic Press the Novare Introductory Physics Program and Video Course.

Nature Notebook: Grades 9-12

Learners may be combined and follow their own interests.

Nature Walks & Scouting: Grades 9-12

Learners may follow their own interests or follow the plan of their local scouting troop or natural history club.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12+	Chemistry Lessons 5 times/week 45 min	HS Chem Intro Introductory Chemistry: A Foundation Michael Faraday's The Chemical History of a Candle The Chemical Elements Coloring and Activity Book
9-12+	Chemistry Labs 1 time/week 60 min	Alveary Chemistry Lab Book

GRADE	SCHEDULE INFO.	BOOKS
9-12	Nature Notebook: Grades 9-12 1+ time/week 20 min+	
9-12	Nature Walks & Scouting: Grades 9-12 1 time/week 30 min+	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Chemistry				
Chemistry Lessons	Chemistry Lessons Nature Walks & Scouting: Grades 1-8	Chemistry Lessons	Chemistry Lessons Nature Notebook: Grades 9-12	Chemistry Lessons Chemistry Labs



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Chemistry Lessons

- ☐ Carve out time to continue or establish the regular habit of spending time in nature, including the use of a nature journal, as appropriate.
- ☐ Obtain materials from the supply lists.
- ☐ Plan for discussion and engagement. Teachers who do not feel that they can discuss the subject of this course with students should plan to watch lesson videos, preread the material, or complete the course alongside their students. Expertise is not necessary, but discussion is as important in math and science as in history and literature.
- ☐ Labs/activities are presented in a fully integrated state in which they are connected to the lessons. To use them in this way, it is helpful for the lab to be scheduled after the 5th lesson. If your lab time is earlier in the week and you will find it difficult to be flexible, plan to delay the start of the lesson schedule so that lab day occurs after the 5th lesson. For example, if your lab time is on Tuesday, you might begin lesson 1 on Wednesday of the first week of school rather than on Monday. Those wanting to use the labs/activities on a more flexible schedule can do so, but should be ready and able to adjust lesson connections.
- ☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.

Term Prep & Teacher Tips

Chemistry Lessons

- ☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
 - 3 small objects from around the house
 - distilled or deionized water
 - mineral spirits/paint thinner (may substitute mineral oil or cooking oil, but won't be able to evaporate the solvent at the end of the lab)
 - 4 recycled soda cans
 - sticky tack, a piece of foam, or anything to hold the thermometer in a soda can and close the opening
 - hydrogen peroxide antiseptic
 - barometer or computer access to check the local weather station
 - ice
 - sugar
 - salt

- spoon
- lemon juice
- stopwatch, such as those found on most electronic devices
- microwave-safe bowl
- microwave
- whisk that you do not care about
- another whisk or electric mixer
- well-ventilated space (porch, carport, etc)
- juice or tea
- quart-sized mason jar with a lid
- jar, cup, or bowl
- slotted spoon, strainer, or coffee filter
- 2 large egg whites (Term 3)
- large mixing bowl
- cookie sheet
- oven
- optional: hard blunt tip (knitting needle, skewer, etc)
- optional: 227g coconut oil from the grocery store, IF you want to use the soap you will make
- optional: lemon-lime soda, ice cream, yogurt, or cottage cheese (Term 3)
- optional: cream of tartar
- optional: piping bag
- optional: food coloring
- optional: parchment paper

Reminders

Chemistry Lessons

☐ Note that egg whites are used in the last lab of the year. Make a note in your calendar if you will need to purchase these closer to the appropriate time. Refer to the lab book for more details.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Science: Chemistry

Chemistry Lessons



HS Chem Intro



Introductory Chemistry: A Foundation



Michael Faraday's The Chemical History of a Candle



The Chemical Elements Coloring and Activity Book

Chemistry Labs



Alveary Chemistry Lab Book



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

Science: Chemistry

-  Alveary High School Chemistry Kit
-  Food-Grade Calcium Chloride
-  Food-Grade Sodium Alginate
-  Food-Grade Sodium Hydroxide
-  Household Items - Science: Chemistry
-  Organic Chemistry Model Kit
-  Scale/Balance

Chemistry Labs

-  Safety Goggles



Quick Links

Science: Chemistry

- ∞ [Chemistry Lab Book](#)

Chemistry Lessons

- ∞ [Alveary Bookshelf](#)
- ∞ [Contact Us for Teachers](#)
- ∞ [Homework Help for HS Science Students](#)
- ∞ [John Muir Laws • Nature Journaling Resources](#)
- ∞ [Khan Academy Video Playlists](#)
- ∞ [Lab Notebook Examples](#)
- ∞ [Seek app from iNaturalist](#)
- ∞ [SkyView Lite for Android](#)
- ∞ [SkyView Lite for iOS](#)

[Click THIS text or scan the QR code for links.](#)



Science: Chemistry

How To Approach



Introduce

- Learners begin each lesson with their existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to consider what learners think about it, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.
- Take note of the chapter or section heading, and consider how the day's topic connects back to prior topic(s).
- Look back as needed. The outline of the book and even of the chapter is helpful in drawing out and connecting ideas. Learners practice making these connections as they proceed through the book.



Read

- Read or do as instructed in the lessons. Teachers should note any Teacher Tips provided.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.
- If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab book further supports major concepts from the text.



Narrate

- Process the ideas of the lesson by explaining a concept, describing an object, retelling events, etc. Consider what the point of the lesson is before deciding what to do with it.
- Learners may use words, pictures, models, graphics, etc, to process and convey ideas in their own way.



Discuss

- Consider together any thoughts, confusion, or concerns about the passage.
- Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher.
- Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.
- Notice if there are any dates to keep for the Book of Centuries or quotes for the Commonplace Book.



Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc, depending on learner

needs and interest.

Science: Chemistry

How To Approach



Introduce

- Before beginning, be sure that the basic rules of lab safety (as listed in the lab manual) are understood and obeyed.
- Every science lab has the same flow, which follows the scientific method and is guided by the lab book.
- Learners begin with an introduction in their lab book. How does this lab activity relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is how hypotheses are formed.
- If learners are unsure what to do with the Introduction, it can help to dialogue with someone as they work through the ideas. Check the Chemistry Companion in the Quick Links if a video tutorial might be helpful. This year is a pilot for this video series, so videos will be added to the Quick Link throughout the year.
- Once they have had a chance to think, learners compose a prelab narration to put these introductory ideas and hypotheses into their lab notebooks. These prelab narrations may seem short and even incomplete if learners are new to keeping a lab notebook, but that is okay.
- If learners have difficulty or are easily frustrated, then consider allowing a digital notebook for typing, a scribe, or use of assistive technology, as appropriate.
- After they complete their prelab narration, learners should collect the listed materials, being sure to let teachers know if something is missing or to remind the teacher if something needs to be purchased at the store.



Lab Procedure

- Learners complete the procedure as instructed in the lab book.
- The lab gives instructions for using the lab notebook to create tables and diagrams.



Analysis & Conclusions

- The last step in the lab is to analyze the data and observations and draw conclusions from them. This postlab narration is prompted in the lab manual. Similar to the prelab narration, the concluding or postlab narration is a chance to think about and put into words. This time, learners are considering what they learned from the lab and what they could learn more about, if they were to continue.
- Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.
- Depending on the interest of the learner and the priorities of the teacher, the learner might be encouraged to spend more time on those ideas of what more they could learn or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.



Term 1

WEEK 1 45m Chemistry Lessons - Lesson 1

Materials: The Chemical History of the Candle Video link

High School Chemistry Lab Book and materials listed within

PREP: Be sure you have read the Course Notes and Planning and Prep. Read Teacher Tip

→ INTRODUCTION

Watch the course introduction video:

∞ Video Link: HS Chemistry Intro (10:02)

In the 1840s, Michael Faraday began a public lecture series for young people, called the Chemical History of the Candle. The lecture series is an excellent demonstration of the scientific method and many fundamental concepts in chemistry. As you watch the lectures this week, create an outline and record any points that stand out to you. This will help you to identify the main points and connect them. You can use these notes to support narration.

→ ACTIVITY

Cut out the spiral from Lab 1 and attach a piece of thread by pushing a threaded needle through the dot in the center; observe the candle, recording observations in your notebook. Light the candle and observe. Using the thread, hold the spiral over the candle (without allowing it to catch fire) and observe. Blow out the candle and observe.

∞ Image Link: Lab Notebook Samples

∞ Video Link: Paper Spiral Demo (1:56)

∞ Image Link: Most Lab 1 materials

→ VIEW, NARRATE, & DISCUSS

Lecture 1

∞ Video Link: Engineering Guy #2 (11:47)

→ ACTIVITY

Complete the follow-up activity in your lab book.

• PLAN WEEKLY

nature walk - record observations

science free read

★ TEACHER TIP

The linked videos are a great way for teachers to keep up with what students are learning, even when they can't read the book alongside students! We especially recommend watching the course introduction video with students!

WEEK 1 45m Chemistry Lessons - Lesson 2

Materials: The Chemical History of the Candle Video link

High School Chemistry Lab Book and materials listed within

Prep: Read Teacher Tip

→ NOTE

Use the prompts bulleted in the lessons to help you engage with the material and to self-assess and decide if you are ready to move on. You don't need to be an expert in any or all of the topics, but the ideas should feel familiar, and you should be able to think about and discuss the ideas.

→ VIEW, NARRATE, & DISCUSS

Lecture 2

∞ Video Link: Engineering Guy #3 (13:44)

- Discuss how Faraday's demo indicates the presence of tiny particles in the candle's flame and how these tiny particles produce the brilliance of the flame.
- What can be concluded about the nature of the particles, and what is still in question regarding combustion production? Consider: Does combustion produce a pure substance or a mixture of substances that can be detected and separated? How did Faraday conclude that the black particles are C? How and what did Faraday observe

★ TEACHER TIP

The discussion prompts bulleted in the lessons provide teachers with helpful ways to engage even when students are working independently.



Term 1

about the invisible particles?

→ ACTIVITY

Complete the follow-up activity in your lab book.

WEEK 1 45m Chemistry Lessons - Lesson 3

Materials: The Chemical History of the Candle Video link

High School Chemistry Lab Book and materials listed within

→ VIEW, NARRATE, & DISCUSS

Lecture 3

∞ Video Link: Engineering Guy #4 (15:42)

- Compare/contrast the three phases observed as the candle burns with the three phases of water.
- Explain why water is still a pure substance even though it can be divided into two parts.
- How do oxygen and hydrogen relate to the combustion reaction?

★ STUDENT TIP

Come to science ready to work a puzzle. If you're missing a piece of that puzzle, don't wait—message Homework Help for Students!

WEEK 1 45m Chemistry Lessons - Lesson 4

Materials: The Chemical History of the Candle Video link

High School Chemistry Lab Book and materials listed within

→ VIEW, NARRATE, & DISCUSS

Lecture 4

∞ Video Link: Engineering Guy #5 (20:00)

- Discuss why it is important that we understand that air is a mixture rather than a pure substance or nothing at all.
- Add to your previous diagram or make a new one, indicating what we now know about the products of the combustion reaction: C soot from the candle, water made of H from the candle, and O from the air, and something else.
- Describe the characteristics of the unknown substance.

→ ACTIVITY

Complete the follow-up activity in your lab book.

WEEK 1 45m Chemistry Lessons - Lesson 5

Materials: The Chemical History of the Candle Video link

High School Chemistry Lab Book and materials listed within

→ VIEW, NARRATE, & DISCUSS

Lecture 5

∞ Video Link: Engineering Guy #6 (21:06)

- How does Faraday demonstrate that the invisible substance is carbon dioxide? How does he show that carbon dioxide is a compounded substance? What is it made of?
- Discuss the uniqueness of carbon-based fuels.
- Describe respiration/breathing in terms of combustion and explain the resulting interdependence of plants and animals.



Term 1

WEEK 1 60m Chemistry Lessons - Lesson 6

60m Chemistry Lab-Lesson 6

Materials: High School Chemistry Lab Book and materials listed within

Observing a Candle

→ LAB DAY

Complete the Analysis and Conclusions for Lab 1. Explore Chapter 3 of the Introductory Chemistry text if you need help.

∞ Video Link: Lab 1 Conclusion (6:37)

WEEK 2 45m Chemistry Lessons - Lesson 7

Materials: Introductory Chemistry

Prep: Read Teacher Tip

→ INTRODUCTION

Just as you did with the Faraday lectures, continue to outline or diagram your readings, noticing main points and recording any definitions or important ideas. Even though you are externalizing the ideas while you do the lesson, you should still narrate and discuss at the end of the lesson. Review your notes at least weekly throughout the year.

Use the video lessons in whatever way is most helpful to you. You can turn on or off closed captions, speed them up or slow them down, use the pause button, etc. Some students prefer to watch the video lesson first and refer to the book as needed, some prefer to read the lesson first and watch the video as a summary or visual recap, and some don't like video lessons and don't use them at all. If you would like additional resources, you can find alternative videos in your Quick Links, as well as the Homework Help link to message Mrs. Merritt.

→ READ, NARRATE, & DISCUSS

Ch.1

∞ Video Link: Scientific Method (6:08)

- What is Chemistry?
- Describe 'the Scientific Method.'
- Discuss the Critical Thinking question in Sec.1-4.
- Discuss Active Learning #6.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

★ TEACHER TIP

The linked videos are a great way for teachers to keep up with what students are learning, even when they can't read the book alongside students! If you prefer a slower, more conceptual approach without the problem-solving and lab tie-in, there is an alternative in the Quick Links.

WEEK 2 45m Chemistry Lessons - Lesson 8

Materials: Introductory Chemistry

→ NOTE

When you complete exercises for the chapter, try a selection from different sections to help discern where your understanding is secure and where you may need some additional review and practice. The problems suggested below are simply those that seemed interesting to us; you may choose different ones. Those demonstrated in the video are noted. The point is not necessarily to get them all correct right away; sometimes exercises can stimulate new thoughts on the ideas. The point is to find where you are comfortable and where you may be confused. If the exercises seem comfortable, then continue. If they seem confusing or too challenging, then review the associated reading(s) and practice some more. This self-assessment is essential as you continue your education. The answers to even-numbered exercises are provided in the back of the book, so that you can check your understanding. Note that books always have some typos; however, don't be too dependent on the answers in the back!



Term 1

→ REVIEW AND PRACTICE

Ch.1 Review and Suggested Exercises:

Act. #12, 13,
1.2 #5, 6,
1.3 #7,
1.4 #10, 13,
1.5 #17

∞ Video Link: Chapter 1 Practice (8:35).

Act. #12, 13,
1.3 #7, edition 8e

WEEK 2 45m Chemistry Lessons - Lesson 9

Materials: Introductory Chemistry

→ NOTE

Always be sure to follow the example problems in the text and complete the self-check exercises as you read. If that means proceeding at a slower pace and spreading lessons over more than one day, that is fine.

If you want to go over zero and negative exponents, consider watching the Math Break video:

∞ Video Link: Math Break: Zero and Negative Exponents (3:48)

→ READ, NARRATE, & DISCUSS

Ch.2:1-2.4

∞ Video Link: Scientific Notation (8:45)

- Why do we use scientific notation, and how does it work?
- What is 'uncertainty' in a measurement, and how do significant figures help?
- Discuss the Critical Thinking question in Sec.2-2.

! STUDENT REMINDER

Remember: scientific thinking is like solving a puzzle. If you're missing a piece of that puzzle, don't wait—message Homework Help for Students!

WEEK 2 45m Chemistry Lessons - Lesson 10

Materials: Introductory Chemistry

→ NOTE

There is a misprint in this section for the 9th edition. The fractions printed in blue are missing their bar/line.

→ READ, NARRATE, & DISCUSS

Ch.2:5-2.6

∞ Video Link: Significant Figures (10:46)

∞ Video Link: Unit Conversions (4:59)

- Discuss Active Learning #7.
- What is a conversion factor? What is its value? Why must this always be true?
- Explain how units help you check your math.

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 2 45m Chemistry Lessons - Lesson 11

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.2:7-2.8

∞ Video Link: Temperature and Density Problems (8:33)



Term 1

- Explain the differences between the three temperature scales.
- What does it mean to say that something is dense?

WEEK 2 60m Chemistry Lessons - Lesson 12

60m Chemistry Lab- Lesson 12

Materials: High School Chemistry Lab Book and materials listed within

Evaluating Density

Prep: Read Teacher Tip

→ LAB DAY

Complete Lab 2.

∞ Video Link: Density Lab (3:55)

∞ Image Link: Examples of Lab 2 materials

★ TEACHER TIP

Asking students to explain what they did in the lab is a great way to begin a discussion! Follow it up with, "Why/how did you do that?"

WEEK 3 45m Chemistry Lessons - Lesson 13

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Ch.2 Review and Suggested Exercises:

Act.#1-5,

2.1 #7, 8,

2.2 #16,

2.3 #22, 24,

2.4 #30, 32,

2.5 #33, 48, 50,

2.6 #64, 66,

2.7 #74,

2.8 #90, 94

∞ Video Link: Chapter 2 Practice (33:07).

Act. #1-4,

2.1 #7, 8,

2.3 #22, 24,

2.5 #33, 48, 50,

2.6 #64,

2.7 #74,

2.8 #90, 94 from 8e

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

☐ nature walk - record observations

☐ science free read

WEEK 3 45m Chemistry Lessons - Lesson 14

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Continue exercises.

- Remember to self-assess! If the exercises made sense, then move on. If they seemed confusing, then go back to the associated readings or use Homework Help.

WEEK 3 45m Chemistry Lessons - Lesson 15

Materials: Introductory Chemistry

High School Chemistry Lab Book

PREP: Read Teacher Tip

★ TEACHER TIP

The videos and discussion prompts are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!



Term 1

→ REVIEW

Briefly review the definition you wrote in the Analysis and Conclusions of Lab 1.

→ READ, NARRATE, & DISCUSS

Ch.4.1-4.4: As you visualize the atom, model it with the parts of your model kit, by sketching it, or with any medium you like, such as modeling clay or Lego!

∞ Video Link: Elements (10:05)

- What is an element?
- Tell what you know about Dalton's theory of atoms.
- What do we mean by the 'law of constant composition'?
- Explain what a chemical formula is and how chemical formulae work.
- Get to know your model kit by building the molecules in example 4.1 (or any suggested in your kit). You can use your kit to help you visualize molecules anytime you want throughout the course.

WEEK 3 45m Chemistry Lessons - Lesson 16

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.4.5-4.7

∞ Video Link: Atoms and Isotopes (8:43)

- What do you know about the parts of atoms?
- How did scientists learn about the parts of the atom?
- Choose at least one Critical Thinking question in Sec.4-5 to discuss.
- What is an isotope?

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 3 45m Chemistry Lessons - Lesson 17

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.4.8-4.9

∞ Video Link: Meeting the Table (5:55)

- Tell about or diagram what you know about the periodic table so far.
- Discuss the differences between different elements, such as gold and hydrogen.
- Are elements atoms or molecules? Explain.

→ SUPPLEMENTAL

Check out some other ways to visualize the periodic table:

∞ Image Link: The Periodic Table in 3D

∞ Image Link: Cylinder with Bulges

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 3 60m Chemistry Lessons - Lesson 18

60m Chemistry - Lesson 18

Materials: Coloring book or any blank periodic table

PREP: Read Teacher Tip

→ ACTIVITY

Use what you have learned to color your own copy of the Periodic Table.

★ TEACHER TIP

Learners encounter a lot more technical content at this disciplinary stage. If they do not remember or fully understand every detail, that is okay. Can they talk about it and explain their thinking? Do they know how to find what they need if they need it? Are they able to lean into the uncertainty with curiosity and a habit of problem-



Term 1

WEEK 4 ☐ 45m Chemistry Lessons - Lesson 19

☐ Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.4.10-4.11

∞ Video Link: Ions (9:42)

- What is an ion?
- How does the periodic table predict ions?
- Tell what you know about ionic compounds.
- Discuss the Critical Thinking question in Sec.4-11.

solving? If you are unsure about their progress, reach out through Contact Us.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 4 ☐ 45m Chemistry Lessons - Lesson 20

☐ Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Ch.4 Review and Suggested Exercises:

Act.#12, 14,

4.1 #4, 5,

4.2 #10-13,

4.4 #20,

4.6 #24, 26, 28,

4.7 #36, 38, 42,

4.8 #44, 48, 49, 53,

4.9 #56, 58, 60, 64,

4.10 #66, 68, 70, 74, 76, 78,

4.11 #80, 82, 84

∞ Video Link: Chapter 4 Practice (13:56).

4.2 #11,

4.4 #20,

4.7 #38, 42,

4.10 #74,

4.11 #84, from 8e

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 4 ☐ 45m Chemistry Lessons - Lesson 21

☐ Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Continue exercises.

- Think you're ready to move on? Remember to take a moment for self-assessment!

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 4 ☐ 45m Chemistry Lessons - Lesson 22

☐ Materials: Coloring book

→ ACTIVITY

If you finished your exercises, then explore the coloring book to begin learning about some elements that you are interested in.



Term 1

WEEK 4 45m Chemistry Lessons - Lesson 23

Materials: Coloring book

Prep: Read Teacher Tip

→ ACTIVITY

Continue exploring elements or choose a game from the back to learn.

★ TEACHER TIP

Ask students to tell you about an element or to teach you a game from the back of the coloring book!

WEEK 4 60m Chemistry Lessons - Lesson 24

60m Chemistry - Lesson 24

Materials: Coloring book

→ ACTIVITY

Continue exploring elements or learn a game from the coloring book.

WEEK 5 45m Chemistry Lessons - Lesson 25

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.5.1-5.2

∞ Video Link: Naming Ionic Compounds (7:33)

- Using a periodic table, how would you name CsBr? Which is the positively charged cation and which is the negatively charged anion? For each, what is their charge, and how many are in the compound?
- Explain the difference between iron (II) chloride and iron (III) chloride. Which is the positively charged cation and which is the negatively charged anion? For each, what is their charge, and how many are in the compound?
- Discuss the Critical Thinking question in Sec.5-2.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 5 45m Chemistry Lessons - Lesson 26

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.5.3-5.4

∞ Video Link: Naming Nonmetal Compounds (4:31)

- How would you name a compound with two nitrogen atoms and four oxygen atoms?
- How would you write the formula for phosphorus pentachloride?
- Use your model kit to build the compounds in example 5.5.

★ TEACHER TIP

The videos and discussion prompts are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 5 45m Chemistry Lessons - Lesson 27

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.5.5-5.7

∞ Video Link: Naming Polyatomic Ions and Acids (8:32)

- What do we mean by "polyatomic ions"?
- In Na₂SO₄, which part is the cation and which part is the anion?

! STUDENT REMINDER

Don't forget to review your lesson notes!



Term 1

- Tell what you know about acids and how we name them.
- Discuss the Critical Thinking question in Sec.5-7.
- Use your model kit to build the polyatomic ions in Table 5.4.

WEEK 5 45m Chemistry Lessons - Lesson 28

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Ch.5 Review and Suggested Exercises:

Act.#4, 6,
5.2 #10, 14, 16,
5.3 #18,
5.5 #30, 34, 36,
5.6 #40,
5.7 #42, 44, 46, 50, 52

∞ Video Link: Chapter 5 Practice (7:20).

Act. #6,

5.7 #50, 52, from 8e

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 5 45m Chemistry Lessons - Lesson 29

Separating Iodine from a Mixture

Materials: High School Chemistry Lab Book and materials listed within

→ READ & NARRATE

Read the Introduction to Lab 3 and complete the introductory narration in your notebook.

∞ Video Link: Separation Lab (9:45)

∞ Image Link: Most Lab 3 materials

WEEK 5 60m Chemistry Lessons - Lesson 30

60m Chemistry Lab- Lesson 30

Materials: High School Chemistry Lab Book and materials listed within

Separating Iodine from a Mixture

PREP: Read Teacher Tip

→ LAB DAY

Complete Lab 3.

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 6 45m Chemistry Lessons - Lesson 31

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Continue exercises.

- Think you're ready to move on? Remember to take a moment for self-assessment!

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



Term 1

WEEK 6 45m Chemistry Lessons - Lesson 32

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ INTRODUCTION

Recall what you have learned so far about the signs of a chemical reaction.

→ READ, NARRATE, & DISCUSS

Ch.6.1-6.2

∞ Video Link: Chemical Reactions (5:35)

- What kinds of observations indicate a chemical reaction?
- Explain the meaning of the following chemical equation: $H_2 + O_2 \rightarrow H_2O$
- Use your model kit to build the reactants for the reaction above. Try to rearrange the atoms to make the product. What do you notice? Try the reaction in section 6.2, if you like.
- What does it mean for a chemical equation to be balanced? Why must it be balanced?

★ TEACHER TIP

The videos and discussion prompts are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 6 45m Chemistry Lessons - Lesson 33

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.6.3: If helpful, use your model kit to build the equation as you read to visualize how it's balanced. Use your kit like this any time you want.

∞ Video Link: Balancing Equations (5:33)

- Explain how to balance a chemical equation.
- Choose at least one Critical Thinking question in Sec.6-3 to discuss.

WEEK 6 45m Chemistry Lessons - Lesson 34

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Ch.6 Review and Suggested Exercises:

Act.#4, 7, 14,

6.1 #1, 6,

6.2 #14, 18, 24, 34,

6.3 #35, 40, 43

∞ Video Link: Chapter 6 Practice (46:34).

Act. #7, 14,

6.1 #1,

6.2 #14, 24, 34,

6.3 #40, 43, from edition 9e

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 6 45m Chemistry Lessons - Lesson 35

Materials: High School Chemistry Lab Book

Link

→ REPORT WRITING

Begin drafting a lab report for the iodine extraction lab. This is simply a formal report based on what is already in your lab notebook. Similar to the linked sample, include a heading with the experiment title, course name, and your name and date; an

! STUDENT REMINDER

Don't forget to review your lesson notes!



Term 1

Introduction (a more polished version of your introductory narration); Methods (this is your procedure or any important information that someone repeating your experiment would need to know); Results (this is your data, including any tables, graphs, or photographs that you want to provide); a Discussion or Data Analysis; and a Conclusion. The last two sections are a more polished version of your analysis and concluding narration.
 ∞ Link: Sample Lab Report

WEEK 6 60m Chemistry Lessons - Lesson 36

60m Chemistry Lab-Lesson 36

Materials: High School Chemistry Lab Book

→ REPORT WRITING

Complete your lab report. Review and edit it to make improvements. While editing, use complete sentences and avoid using the first person. In technical writing, such as this, the sentence "I tied the string to the end of the meter stick." should be rephrased, "The string was tied to the end of the meter stick."

∞ Image Link: Use Passive Voice

WEEK 7 45m Chemistry Lessons - Lesson 37

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Continue exercises.

- Think you're ready to move on? Remember to take a moment for self-assessment!

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 7 45m Chemistry Lessons - Lesson 38

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.7.1-7.2

∞ Video Link: Precipitation Reactions (6:08)

- Describe some factors that cause a reaction to tend to occur. Do you think these are connected to the signs of a reaction in the last chapter?
- Describe what happens in a precipitation reaction, such as when silver nitrate and potassium chloride are mixed in aqueous solution. Use a sketch or diagram, if helpful.
- Why is the formation of precipitate evidence of a chemical reaction? Use a sketch or diagram, if helpful.
- Discuss the Critical Thinking question in Sec.7-2.

★ TEACHER TIP

The videos and discussion prompts are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 7 45m Chemistry Lessons - Lesson 39

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.7.3-7.4

∞ Video Link: Acid-Base Reactions (Reactions Producing Liquid Water) (6:13)

! STUDENT REMINDER

Don't forget to review your lesson notes!



Term 1

- What are acids and bases?
- What do we mean by "strong" acids and bases?
- Explain what is produced when strong acids and bases react?

WEEK 7 45m Chemistry Lessons - Lesson 40

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.7.5

∞ Video Link: Redox Reactions (Reactions Transferring Electrons) (4:00)

- What do we mean by "oxidation-reduction"? Use a sketch or diagram, if helpful.
- If an element is one of the products or reactants, then the reaction is oxidation-reduction. Discuss this statement.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 7 45m Chemistry Lessons - Lesson 41

Observing Microscale Reactions

Materials: High School Chemistry Lab Book and materials listed within

→ READ & NARRATE

Read the Introduction to Lab 4 and complete the introductory narration in your notebook.

∞ Video Link: Microscale Reactions Lab (5:22)

∞ Image Link: Most Lab 4 materials

WEEK 7 60m Chemistry Lessons - Lesson 42

60m Chemistry Lab - Lesson 42

Materials: High School Chemistry Lab Book and materials listed within

Observing Microscale Reactions

PREP: Read Teacher Tip

→ LAB DAY

Complete Lab 4.

★ TEACHER TIP
Ask students to explain their lab to you!

WEEK 8 45m Chemistry Lessons - Lesson 43

Materials: Introductory Chemistry

→ NOTE

These sections are about classifying the reactions we have been talking about. The new terms are something to think about, but do not spend time memorizing different types of reactions. They are more for your awareness.

→ READ, NARRATE, & DISCUSS

Ch.7.6-7.7

- Look back at the objectives for these sections. Discuss the ideas.
- Discuss the Critical Thinking question in Sec.7-7.

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

- **PLAN WEEKLY**
 - ☐ nature walk - record observations
 - ☐ science free read

WEEK 8 45m Chemistry Lessons - Lesson 44

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Ch.7 Review and Suggested Exercises:



Term 1

Act.#2, 4, 14,
7.1 #2,
7.2 #6, 10, 18, 22,
7.3 #24, 28,
7.4 #32, 36, 40,
7.5 #44, 46, 50,
7.6 #52, 54,
7.7 #60, 62

∞ Video Link: Chapter 7 Practice (29:38).

Act. #2,
7.2 #18,
7.3 #28,
7.4 #40,
7.6 #54, from 9e

WEEK 8 45m Chemistry Lessons - Lesson 45

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Continue exercises.

- How is your self-assessment going? If you need help, click on Homework Help!

! STUDENT REMINDER
Don't forget to review your lesson notes!

WEEK 8 45m Chemistry Lessons - Lesson 46

Materials: Introductory Chemistry

Prep: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.8.1-8.2

∞ Video Link: Atomic Mass Units (4:41)

→ NOTICE THAT THE DISCUSSION PROMPTS COME FROM THE OBJECTIVES AT THE START OF ASSIGNED SECTIONS. THESE OBJECTIVES HELP YOU KNOW WHAT YOU SHOULD BE ABLE TO NARRATE AND DISCUSS. YOU DO NOT NEED TO HAVE FORMULAE AND PROCEDURES MEMORIZED, BUT RATHER BE ABLE TO DISCUSS THEM. IF THE IDEAS SEEM FAMILIAR ENOUGH TO NARRATE AND DISCUSS, THEN CONTINUE. IF THEY SEEM CONFUSING OR TOO CHALLENGING, THEN REVIEW THE ASSOCIATED READING(S) OR SEND A MESSAGE TO HOMEWORK HELP. THESE SELF-ASSESSMENT SKILLS WILL HELP YOU AS YOU PROGRESS IN YOUR EDUCATION.

- What do we mean by "average mass"? Explain how this idea can be used to count something.
- Explain atomic mass. How is it determined?
- What are "atomic mass units"?

★ TEACHER TIP
Note that the discussion prompts bulleted in the lessons come from the objectives at the start of the assigned sections. These objectives provide teachers with helpful ways to engage, even when students are working independently.

WEEK 8 45m Chemistry Lessons - Lesson 47

Materials: Introductory Chemistry

Website link

→ READ, NARRATE, & DISCUSS

Ch.8.3-8.4

∞ Video Link: The Mole (4:22)

- Use the idea of Avogadro's number to explain what a mole is.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!



Term 1

- Explain how one converts between moles, mass, and number of atoms. Diagram or demonstrate, as needed.
 - Tell about the process of problem-solving. How do you do it?
 - Choose at least one Critical Thinking question in Sec.8-3 to discuss.
 - Did you know that some unit definitions have recently changed? The values have not changed much, but the way they are standardized has been updated. Check out the reasons and the changes at The New SI link.
- ∞ Website Link: The New SI

WEEK 8 60m Chemistry Lessons - Lesson 48

60m Chemistry Lab - Lesson 48

Materials: High School Chemistry Lab Book and materials listed within

Counting by Weighing

PREP: Read Teacher Tip

→ LAB DAY

Complete Lab 5.

∞ Video Link: Counting by Weighing Lab (2:04)

∞ Image Link: Most Lab 5 materials

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 9 45m Chemistry Lessons - Lesson 49

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.8.5-8.6

∞ Video Link: Molar Mass and Percent Composition (9:58)

- What is molar mass?
- Explain how one converts between moles and mass of a sample of a chemical compound. Diagram or demonstrate, as needed.
- What do we mean by "mass percent"?

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 9 45m Chemistry Lessons - Lesson 50

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.8.7-8.9: If helpful, use your model kit to collect the correct ratio of atoms indicated, to visualize multiples of this ratio, and to experiment with how the atoms might fit together.

∞ Video Link: Empirical and Molecular Formulae (8:15)

- What do we mean by "empirical formula"?
- Discuss how empirical formulae are determined.
- How can an empirical formula help us get to a molecular formula?
- Discuss the Critical Thinking question in Sec.8-8.

★ TEACHER TIP

The videos and discussion prompts are a great way to keep up and engage with students even when they are working independently. Your positive support matters!

WEEK 9 45m Chemistry Lessons - Lesson 51

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Ch.8 Review and Suggested Exercises:

Act. #4, 17,

! STUDENT REMINDER

Don't forget to review your lesson notes!



Term 1

8.2 #4, 6, 8,
8.3 #10, 12, 18, 24,
8.5 #26, 30, 34, 38,
8.6 #48,
8.7 #54,
8.8 #60, 66,
8.9 #76, 78

∞ Video Link: Chapter 8 Practice (40:35).

Act. #4, 17,

8.2 #8,

8.3 #18,

8.5 #34, from 9e

WEEK 9 45m Chemistry Lessons - Lesson 52

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Continue exercises.

- Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 9 45m Chemistry Lessons - Lesson 53

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.9.1-9.2: If helpful, use the model kit to build the water molecules on the reactant side of the equation. Use only those atoms to build the products. Use the model kit for any of the reactions you want to visualize in this way.

∞ Video Link: Stoichiometry Part 1 (3:03)

- What kind of information does a chemical equation provide?
- How do we relate moles of reactants and products using a chemical equation?

WEEK 9 60m Chemistry Lessons - Lesson 54

60m Chemistry Lab - Lesson 54

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.9.3: If helpful, use the model kit to represent the propane and oxygen reactants in example 9.4. Use only those atoms and all of those atoms to try to build the products. Do you have too much or too little of something? How might you resolve the discrepancy? Use the model kit for any of the reactions you want to visualize in this way.

∞ Video Link: Stoichiometry Part 2 (5:42)

- How do we relate masses of reactants and products in a chemical equation?
- Discuss the Critical Thinking question in Sec.9-3.

WEEK 10 45m Chemistry Lessons - Lesson 55

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.9.4-9.5

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.



Term 1

∞ Video Link: Limiting Reagents (6:54)

- What do we mean by "limiting reagent" or "limiting reactant"?
- Explain how one can identify a limiting reagent?
- How does a limiting reagent affect stoichiometric calculations?

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 10 ☐ 45m Chemistry Lessons - Lesson 56

☐ Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.9.6

∞ Video Link: Percent Yield (2:52)

- Discuss actual yield versus theoretical yield.
- Explain how actual yield can be determined from theoretical yield.

→ REVIEW AND PRACTICE

Ch.9 Review and Suggested Exercises:

Act. #2, 6,

9.1 #2, 4,

9.2 #12, 16,

9.3 #19, 21, 30, 38,

9.5 #52, 57,

9.6 #62, 65

∞ Video Link: Chapter 9 Practice (31:01).

Act. #2, 6,

9.3 #19, 21,

9.5 #57,

9.6 #65, from 9e

★ TEACHER TIP

The videos and discussion prompts are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 10 ☐ 45m Chemistry Lessons - Lesson 57

☐ Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Continue exercises.

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 10 ☐ 45m Chemistry Lessons - Lesson 58

☐ Materials: Current Events link

Introductory Chemistry

→ READ, NARRATE, & DISCUSS

What interests you? Find out what chemistry can teach you about that interest by searching for it or browse to discover something new at one of the current event resources below:

∞ Website Link: Science News for Students

∞ Website Link: Chemistry at Science Daily

∞ Website Link: Chemical & Engineering News

→ REVIEW AND PRACTICE

Continue exercises.

- Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!



Term 1

WEEK 10 45m Chemistry Lessons - Lesson 59

Evaluating Limiting Reagents

Materials: High School Chemistry Lab Book and materials listed within

→ READ & NARRATE

Read the Introduction to Lab 6 and complete the introductory narration in your notebook.

∞ Video Link: Limiting Reagents Lab (1:10)

∞ Image Link: Most Lab 6 materials

WEEK 10 60m Chemistry Lessons - Lesson 60

60m Chemistry Lab - Lesson 60

Materials: High School Chemistry Lab Book and materials listed within

Evaluating Limiting Reagents

PREP: Read Teacher Tip

→ LAB DAY

Complete Lab 6.

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 11 45m Chemistry Lessons - Lesson 61

Materials: High School Chemistry Lab Book

Introductory Chemistry

→ REPORT WRITING & REVIEW

Spend this week writing a lab report for Lab 6, then review what you have learned so far this term.

- Review and edit the lab report before turning it in. Use complete sentences, good grammar, third-person voice, and significant figures and scientific notation, as appropriate.

- As you look back over your notes and narrate objectives from the chapters, reflect on whether ideas make sense or seem confusing. If anything is confusing, read back over the associated section(s), try some exercises, and/or use Homework Help.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 11 45m Chemistry Lessons - Lesson 62

Materials: High School Chemistry Lab Book

Introductory Chemistry

PREP: Read Teacher Tip

→ REPORT WRITING & REVIEW

Spend this week writing a lab report for Lab 6, then review what you have learned so far this term.

★ TEACHER TIP

When student(s) take exams next week, evaluate what they recall, but more importantly whether they can explain their thinking and demonstrate a habit of problem-solving. If no relational growth is observed, consider why: do they need more practice, more support, less business, etc? Reach out through Contact Us, if you need help.

WEEK 11 45m Chemistry Lessons - Lesson 63

Materials: High School Chemistry Lab Book

Introductory Chemistry

→ REPORT WRITING & REVIEW



Term 1

Spend this week writing a lab report for Lab 6, then review what you have learned so far this term.

WEEK 11 45m Chemistry Lessons - Lesson 64

Materials: High School Chemistry Lab Book

Introductory Chemistry

→ REPORT WRITING & REVIEW

Spend this week writing a lab report for Lab 6, then review what you have learned so far this term.

WEEK 11 45m Chemistry Lessons - Lesson 65

Materials: High School Chemistry Lab Book

Introductory Chemistry

→ REPORT WRITING & REVIEW

Spend this week writing a lab report for Lab 6, then review what you have learned so far this term.

WEEK 11 60m Chemistry Lessons - Lesson 66

60m Chemistry Lab - Lesson 66

Materials: High School Chemistry Lab Book

Introductory Chemistry

→ REPORT WRITING & REVIEW

Spend this week writing a lab report for Lab 6, then review what you have learned so far this term.

WEEK 12 45m Chemistry Lessons - Lesson 67

Exam Week

→ EXAMS

Answer question(s) related to course. Complete about three questions per day or as directed by your teacher. You may use the reference material inside the covers of your book as needed.

WEEK 12 45m Chemistry Lessons - Lesson 68

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Chemistry Lessons - Lesson 69

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Chemistry Lessons - Lesson 70

Exam Week

→ EXAMS

Answer question(s) related to course.



Term 1

WEEK 12 45m Chemistry Lessons - Lesson 71

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 60m Chemistry Lessons - Lesson 72

60m Chemistry Lab - Lesson 72

Exam Week

→ EXAMS

Answer question(s) related to course.



Term 2

WEEK 1 45m Chemistry Lessons - Lesson 73

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ REMINDER

Use the video lessons in whatever way is most helpful to you. You can turn on or off closed captions, speed them up or slow them down, use the pause button, etc. Some students prefer to watch the video lesson first and refer to the book as needed, some prefer to read the lesson first and watch the video as a summary or visual recap, and some don't like video lessons and don't use them at all. If you would like additional resources, you can find alternative videos in your Quick Links, as well as the Homework Help link to message Mrs. Merritt.

→ READ, NARRATE, & DISCUSS

Ch.10.1-10.3

∞ Video Link: Energy and Heat (7:39)

- Discuss "energy," types of energy, and conservation of energy.
- Explain the difference between temperature and heat.
- Discuss the direction of energy flow in a system.
- Discuss the Critical Thinking question in Sec.10-1.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

★ TEACHER TIP

The linked videos and the objectives at the start of each section are a great way for teachers to keep up with what students are learning, even when they can't read the book alongside students! If you prefer a slower, more conceptual approach without the problem-solving and lab tie-in, there are some alternatives in the Quick Links.

WEEK 1 45m Chemistry Lessons - Lesson 74

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.10.4-10.5

∞ Video Link: Thermodynamics (8:24)

- Explain how the internal energy (E) of a system is affected when heat flows into the system. When heat flows out of the system?
- Discuss the measurement of heat.
- Discuss the Critical Thinking question in Sec.10-4.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 1 45m Chemistry Lessons - Lesson 75

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.10.6-10.7

∞ Video Link: Enthalpy (4:05)

→ RECALL THAT THESE DISCUSSION PROMPTS COME FROM THE OBJECTIVES AT THE START OF ASSIGNED SECTIONS. IN THE NEXT FEW WEEKS, THE PROMPTS RELATED TO THOSE OBJECTIVES WON'T BE IN YOUR LESSON PLANS, SO PRACTICE USING THEM IN THE TEXT FOR YOUR SELF-ASSESSMENT. IF THE IDEAS SEEM FAMILIAR ENOUGH TO NARRATE AND DISCUSS, THEN CONTINUE. IF THEY SEEM CONFUSING OR TOO CHALLENGING, THEN REVIEW THE ASSOCIATED READING(S) OR SEND A MESSAGE TO HOMEWORK HELP.

- Explain what you know about "enthalpy."
- Explain Hess's Law, relating to enthalpy and reactions that occur in steps.
- Discuss the Critical Thinking question in Sec.10-7.



Term 2

WEEK 1 45m Chemistry Lessons - Lesson 76

Materials: Introductory Chemistry

→ CATCH UP & REVIEW

Complete any unfinished work, then do a cumulative review of all you have learned, including what you learned in Term 1.

WEEK 1 45m Chemistry Lessons - Lesson 77

Finding Enthalpy

Materials: High School Chemistry Lab Book and materials listed within

→ READ & NARRATE

Read the Introduction to Lab 7 and complete the introductory narration in your notebook.

∞ Video Link: Enthalpy Lab (3:50)

∞ Image Link: Most Lab 7 materials

WEEK 1 60m Chemistry Lessons - Lesson 78

60m Chemistry Lab - Lesson 78

Materials: High School Chemistry Lab Book and materials listed within

Finding Enthalpy

PREP: Read Teacher Tip

→ LAB DAY

Complete Lab 7.

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 2 45m Chemistry Lessons - Lesson 79

Materials: Introductory Chemistry

Prep: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.10.8-10.9

∞ Video Link: Energy and Us (4:55)

- What is meant by the "quality" of energy? How does this change as energy is used?
- Discuss what you know about types of energy resources.
- Discuss the Critical Thinking question in Sec.10-9.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

★ TEACHER TIP

Use the objectives at the start of assigned sections to engage in discussion and note progress even when students are working independently. Students should begin using those objectives to assess their learning over the next few weeks, if they aren't already doing so.

WEEK 2 45m Chemistry Lessons - Lesson 80

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.10.10

∞ Video Link: Entropy (2:43)

- Discuss energy as a "driving force" for natural processes.

→ REVIEW AND PRACTICE

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.



Term 2

Ch.10 Review and Suggested Exercises:

Act. #2, 9,
10.2 #8,
10.3 #12, 14,
10.4 #17, 20,
10.5 #22, 24, 30, 32,
10.6 #42,
10.7 #48,
10.9 #58,
10.10 #64

∞ Video Link: Chapter 10 Practice (21:22).

Act. #9,
10.4 #17, 20,
10.5 #24, 32,
10.6 #42,
10.7 #48, from 9e

WEEK 2 45m Chemistry Lessons - Lesson 81

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Continue exercises.

- Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 2 45m Chemistry Lessons - Lesson 82

Materials: Current Events link

→ INTRODUCTION

Not all chemists agree about the role that entropy plays in the universe. Physical Chemist Arie Ben-Naim describes entropy as "one of the most misunderstood concepts in physics" and argues that scientists "do not have license to predict the death of the universe..." You will have the opportunity to learn much more about entropy in future coursework. While scientists continue to research and strive to understand thermodynamics better, we can still seek ways to use energy responsibly and steward well all that we have been given. The linked article, "Energy Current Event," will give you one example of this complex challenge.

→ READ, DISCUSS, & NARRATE

∞ Article Link: Energy Current Event

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 2 45m Chemistry Lessons - Lesson 83

Materials: Internet or library access

PREP: Read Teacher Tip

→ INTRODUCTION

As citizens, we need to think about how to solve problems, like how we generate and use energy. These kinds of problems can be difficult because there are rarely good or bad solutions. We want to make decisions that do the most good and the least harm, and sometimes, the best solution may be different in different communities. You will have three periods for this research and will prepare a composition about your findings later.

★ TEACHER TIP

Ask students about their citizen science project. What are they researching? How are they planning to present their findings?



Term 2

→ RESEARCH

Learn about the life cycle of one traditional energy source in your area, such as fossil fuels, and one alternative energy source, such as solar, wind, nuclear, or fuel cell technology. For each energy source, try to answer these questions:

- What raw materials are needed to support the technology? How and from where are those materials obtained?
- How is the energy produced and distributed?
- What kind of waste is produced by using this energy source?
- How long do the technology and its support structures last? What kind of waste needs to be disposed of?
- How does this energy source positively and negatively affect people and the environment?
- What systems in society support the use of this energy source (e.g., economic, transportation, etc)?

WEEK 2 60m Chemistry Lessons - Lesson 84

60m Chemistry Lab - Lesson 84

Materials: Internet or library access

→ RESEARCH

Continue researching energy sources.

WEEK 3 45m Chemistry Lessons - Lesson 85

Materials: Internet or library access

→ RESEARCH

Complete your research. Use any remaining time to begin thinking about your composition. Effective communication of complex ideas is a major challenge in science, so think about how you communicate best and plan your composition thoughtfully! You could compose an expository essay, a PowerPoint or poster presentation, a graphic display using Canva or watercolor, or any other form of expression that allows you to communicate what you have learned effectively.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 3 45m Chemistry Lessons - Lesson 86

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.11.1-11.4: Just like in Ch.4, model what you are learning about the atom, so that you can visualize it.

∞ Video Link: Light Energy (3:49)

- Discuss Rutherford's nuclear model of the atom.
- Tell what you know about electromagnetic radiation.
- Explain how atoms emit light.
- Discuss how emission spectra demonstrate that light is quantized. Use a diagram, if helpful.
- Discuss the Critical Thinking question in Sec.11-4.

★ TEACHER TIP

The videos and discussion prompts/section objectives are a great way to keep up and engage with students even when they are working independently. Your positive support matters!

WEEK 3 45m Chemistry Lessons - Lesson 87

Materials: Introductory Chemistry

Website link

→ READ, NARRATE, & DISCUSS

! STUDENT REMINDER

Don't forget to review your lesson notes!



Term 2

Ch.11.5-11.7

∞ Video Link: Updating the Atom (best with full screen) (6:04)

- Discuss Bohr's model with circular orbits. Sketch or diagram, if helpful.
- Explain what you know about the wave mechanical model. Sketch or diagram, if helpful.
- Tell what you know about s, p, and d orbital shapes. Sketch or diagram, if helpful.
- For a 3D visualization of the orbitals described in section 11.7, check out "hydrogenic atomic orbitals" at the link. You can manipulate and rotate them to see them better.
- ∞ Website Link: Orbital Visualizations

WEEK 3 45m Chemistry Lessons - Lesson 88

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.11.8-11.9

∞ Video Link: Electron Configuration (best with full screen) (9:00)

- Tell what you know about energy levels and orbitals in the wave mechanical model. Sketch or diagram, if helpful.
- Tell what you know about electron spin. Sketch or diagram, if helpful.
- Tell what you know about how electrons fill energy levels in atoms larger than hydrogen. Sketch or diagram, if helpful.
- Discuss valence versus core electrons.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 3 45m Chemistry Lessons - Lesson 89

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.11.10

∞ Video Link: Electron Configuration Continued (6:38)

- Explain what you know about electron configuration in atoms with atomic numbers greater than 18.
- Discuss the Critical Thinking question in Sec.11-10.

WEEK 3 60m Chemistry Lessons - Lesson 90

60m Chemistry Lab - Lesson 90

Materials: High School Chemistry Lab Book and materials listed within

Observing the Chemistry of Electronic Configuration

PREP: Read Teacher Tip

→ LAB DAY

Complete Lab 8.

∞ Video Link: Electron Configuration Lab (2:01)

∞ Image Link: Most Lab 8 materials

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 4 45m Chemistry Lessons - Lesson 91

Materials: Introductory Chemistry

Coloring book

→ READ, NARRATE, & DISCUSS

Ch.11.11

∞ Video Link: Periodic Trends (4:06)

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY



Term 2

- Using a periodic table, describe trends in atomic properties.

→ ACTIVITY

Color additional copies of the Periodic Table to illustrate the periodic trends you have learned about.

- ☐ nature walk - record observations
- ☐ science free read

WEEK 4 ☐ 45m Chemistry Lessons - Lesson 92

☐ Materials: Introductory Chemistry

Prep: Read Teacher Tip

→ NOTE

Now that you've got the hang of using the chapter exercises to help self-assess your learning, you should choose some exercises for yourself. Don't forget to consider whether they feel comfortable or confusing before deciding whether to move on or practice more. Remember that the point is not necessarily to get them all correct right away; sometimes exercises can stimulate new thoughts on the ideas. The point is to find where you are comfortable and where you may be confused.

→ REVIEW AND PRACTICE

Ch.11 Review and Suggested Exercises:

11.1 #2,
choose 2 from 11.2,
choose 1 from 11.3,
choose 2 from 11.4,
11.6 #26,
choose 3 from 11.7,
11.9 #50, 54,
11.10 #60, 62, 66,
choose 2 from 11.11

∞ Video Link: Chapter 11 Practice (15:59).

11.9 #50, 54,
11.10 #60, 62, 66, from 9e

★ TEACHER TIP

Students will begin selecting some of their own exercises to practice this week (if they haven't already done so). Check and see how they are doing!

WEEK 4 ☐ 45m Chemistry Lessons - Lesson 93

☐ Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Continue exercises.

- Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 4 ☐ 45m Chemistry Lessons - Lesson 94

☐ Materials: Introductory Chemistry

Prep: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.12.1-12.3

∞ Video Link: Chemical Bonding (6:17)

★ TEACHER TIP

Note that students should use the objectives from the start of the assigned sections to assess their learning. Check that they are doing so and encourage them to message Homework Help, if they need support.



Term 2

- Looking at the objectives at the start of the assigned sections, tell what you know about each topic.
- Discuss the Critical Thinking question in Sec.12-2.

WEEK 4 45m Chemistry Lessons - Lesson 95

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.12.4-12.5

∞ Video Link: Understanding Ionic Solids (3:24)

- Looking at the objectives at the start of the assigned sections, tell what you know about each topic.
- Discuss the Critical Thinking question in Sec.12-5.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 4 60m Chemistry Lessons - Lesson 96

60m Chemistry Lab - Lesson 96

Materials: Notebook or computer to compose

→ REPORT WRITING

Prepare a composition to report on your energy research. This could be an expository essay, a PowerPoint or poster presentation, a graphic display using Canva or watercolor, or any other form of expression that allows you to effectively communicate what you have learned.

WEEK 5 45m Chemistry Lessons - Lesson 97

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.12.6

∞ Video Link: Lewis Structures (4:26)

- Looking at the objectives at the start of the assigned section, tell what you know about the topic.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 5 45m Chemistry Lessons - Lesson 98

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.12.7

∞ Video Link: Lewis Structures Continued (5:42)

- Tell what you know about the objectives at the start of this section.

★ TEACHER TIP

Don't forget to check that students are using the objectives at the start of assigned sections to assess their learning. Encourage them to message Homework Help, if they need support.

WEEK 5 45m Chemistry Lessons - Lesson 99

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.12.8-12.9: If helpful, use the model kit to assemble some of the structures and

★ TEACHER TIP

Learners encounter a lot more technical content at this disciplinary stage. If they do not remember or fully understand every detail, that is okay. Can they talk about it and explain their thinking?



Term 2

observe their 3-dimensional geometry.

∞ Video Link: Understanding Covalent Structures (4:42)

- Tell what you know about the objectives at the start of these sections.

Do they know how to go back and find what they need if they need it? Are they able to lean into the uncertainty with curiosity and a habit of problem-solving? If you are unsure about their progress, reach out through Contact Us.

WEEK 5 45m Chemistry Lessons - Lesson 100

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.12.10: If helpful, use the model kit to assemble some of the structures and observe their 3-dimensional geometry. You can try them first with single bonds and then compare them to the geometry when you add the double bond.

∞ Video Link: VSEPR Bonus (2:02)

- Tell what you know about the objectives at the start of this section.

→ REVIEW AND PRACTICE

Ch.12 Review and Suggested Exercises:

12.1 #6,

choose 3 from 12.2,

choose 2 from 12.3,

choose 3 from 12.4,

12.5 #46,

choose 3 from 12.6-7,

choose 3 from 12.9-10

∞ Video Link: Chapter 12 Practice (18:48).

12.3 #22,

12.5 #46,

12.6-7 #60, 66,

12.9-10 #82, from 9e

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 5 45m Chemistry Lessons - Lesson 101

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Continue exercises.

- Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

WEEK 5 60m Chemistry Lessons - Lesson 102

60m Chemistry Lab - Lesson 102

Materials: Introductory Chemistry

→ CATCH UP & REVIEW

Complete any unfinished work, then do a cumulative review of all you have learned. As you look back over your notes and narrate objectives from the chapters, reflect on whether the ideas make sense or seem confusing. If anything is confusing, read back over the associated section(s), try some exercises, and/or use Homework Help.



Term 2

WEEK 6 45m Chemistry Lessons - Lesson 103

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.13.1-13.2

∞ Video Link: Boyle's Law (5:16)

- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 6 45m Chemistry Lessons - Lesson 104

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.13.3

∞ Video Link: Charles' Law (3:52)

- Tell what you know about the objectives at the start of this section.
- Discuss the Critical Thinking question in Sec.13-3.

WEEK 6 45m Chemistry Lessons - Lesson 105

Materials: Introductory Chemistry

→ NOTE

There is a typo in example 13.7 in edition 8. Avogadro's Law should have "n" in the denominator rather than "T."

→ READ, NARRATE, & DISCUSS

Ch.13.4

∞ Video Link: Avogadro's Law (2:31)

- Tell what you know about the objectives at the start of this section.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 6 45m Chemistry Lessons - Lesson 106

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.13.5

∞ Video Link: Ideal Gas Law (3:05)

- Tell what you know about the objectives at the start of this section.

★ TEACHER TIP

Don't forget to check that students are using the objectives at the start of assigned sections to assess their learning. Encourage them to message Homework Help, if they need support.

WEEK 6 45m Chemistry Lessons - Lesson 107

Determining the Peroxide Content of a Commercial Antiseptic

Materials: High School Chemistry Lab Book and materials listed within

→ READ & NARRATE

Read the Introduction to Lab 9 and complete the introductory narration in your

! STUDENT REMINDER

Don't forget to review your lesson notes!



Term 2

notebook.
 ∞ Video Link: Peroxide Lab (2:41)
 ∞ Image Link: Most Lab 9 materials

WEEK 6 60m Chemistry Lessons - Lesson 108

60m Chemistry Lab - Lesson 108

Materials: High School Chemistry Lab Book and materials listed within

Determining the Peroxide Content of a Commercial Antiseptic

PREP: Read Teacher Tip

→ LAB DAY

Complete Lab 9.

★ TEACHER TIP
 Ask students to explain their lab to you!

WEEK 7 45m Chemistry Lessons - Lesson 109

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.13.6-13.7

∞ Video Link: Dalton's Law of Partial Pressures (7:19)

- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER
 Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 7 45m Chemistry Lessons - Lesson 110

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.13.8-13.10

∞ Video Link: Kinetic Theory (7:05)

- Tell what you know about the objectives at the start of these sections.
- Choose at least one Critical Thinking question in Sec.13-8 or 13-10 to discuss.

★ TEACHER TIP
 The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 7 45m Chemistry Lessons - Lesson 111

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Ch.13 Review. Then choose exercises to complete as in previous chapters. Remember to try a selection of exercises to help discern where your understanding is secure and where you may need some additional review and practice. If the exercises seem comfortable, then continue. If they seem confusing or too challenging, review the associated reading(s) and practice more. The answers to even-numbered exercises are provided at the back of the book so you can check your understanding.

∞ Video Link: Chapter 13 Practice (41:23).

- 13.1 #12,
- 13.2 #22,
- 13.3 #36,
- 13.4 #42,

! STUDENT REMINDER
 Don't forget to review your lesson notes!



Term 2

13.5 #52, 58,
13.6 #70, 74,
13.10 #96, 100, from 9e

WEEK 7 45m Chemistry Lessons - Lesson 112

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Continue exercises.

- Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 7 45m Chemistry Lessons - Lesson 113

Materials: High School Chemistry Lab Book

→ REPORT WRITING

Begin writing a formal lab report for the candle enthalpy lab. It's been a while since you completed that lab, so it may be a challenge if you didn't keep detailed lab records. If this is the case, think about what would make your lab record more useful to you now and note this in your report.

WEEK 7 60m Chemistry Lessons - Lesson 114

60m Chemistry Lab - Lesson 114

Materials: High School Chemistry Lab Book

→ REPORT WRITING

Complete your lab report. Review and edit it to make improvements. Use complete sentences, good grammar, third-person voice, and significant figures and scientific notation, as appropriate. As you read your report, imagine you are another scientist unfamiliar with the experiment. Does the report make sense? Would you be able to repeat the experiment using the lab report? If not, edit the report to make it so.

WEEK 8 45m Chemistry Lessons - Lesson 115

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.14.1-14.2

∞ Video Link: Heats of Fusion and Vaporization (7:55)

- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 8 45m Chemistry Lessons - Lesson 116

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.14.3-14.4: If helpful, use the model kit to build water molecules. Then visualize water's intermolecular forces in solid, liquid, and gas states. What happens when salt ions are added to the liquid water model? Can you visualize it?

∞ Video Link: London Dispersion and Vapor Pressure (4:36)

★ TEACHER TIP

Don't forget to check that students are using the objectives at the start of assigned sections to assess their learning. Encourage them to message Homework Help, if they need support.



Term 2

- Tell what you know about the objectives at the start of these sections.
- Discuss the Critical Thinking question in Sec.14-3.

WEEK 8 45m Chemistry Lessons - Lesson 117

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.14.5-14.6

∞ Video Link: Crystalline Solids (3:37)

- Try building the crystalline solids in Figure 14.14 with your model kit.
- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 8 45m Chemistry Lessons - Lesson 118

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Ch.14 Review. Then choose exercises to complete.

∞ Video Link: Chapter 14 Practice (18:41).

Act. #13, 17, 19,

14.1 #6 (also #8, Act. #20, 22),

14.2 #16,

14.3 #28, 30,

14.4 #38,

14.5 #46, from 9e

- Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

! STUDENT REMINDER
Don't forget to review your lesson notes!

WEEK 8 45m Chemistry Lessons - Lesson 119

Separating a Mixture of Pigments

Materials: High School Chemistry Lab Book and materials listed within

→ READ & NARRATE

Read the Introduction to Lab 10 and complete the introductory narration in your notebook.

∞ Video Link: Chromatography Lab (2:38)

∞ Image Link: Most Lab 10 materials

★ TEACHER TIP
Ask students to explain their lab to you!

WEEK 8 60m Chemistry Lessons - Lesson 120

60m Chemistry Lab - Lesson 120

Materials: High School Chemistry Lab Book and materials listed within

Separating a Mixture of Pigments

PREP: Read Teacher Tip

→ LAB DAY

Complete Lab 10.



Term 2

WEEK 9 45m Chemistry Lessons - Lesson 121

Materials: Website link

→ READ & PREPARE

Explore the linked citizen science project you will participate in next term and prepare by discussing it with your teacher. You will use the nitrate and pH strips in your lab kit, as well as observe any stream invertebrates you find. Talk to your teacher about where you will collect your data for the project and make any necessary plans. Be sure to have a way to collect your water samples and possibly a dip net.

- ∞ Website Link: Clean Water Hub
- ∞ Website Link: Macroinvertebrate Gallery
- ∞ Website Link: Macroinvertebrate ID Guide

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 9 45m Chemistry Lessons - Lesson 122

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.15.1-15.2: If helpful, use the model kit to build water molecules and demonstrate what happens when salt ions are added to the liquid water model. Build any of the molecules in Figures 15.3-5 and see how they interact with a water molecule.

- ∞ Video Link: Solutions (3:45)

- Tell what you know about the objectives at the start of these sections.

★ TEACHER TIP

Don't forget to check that students are using the objectives at the start of assigned sections to assess their learning. Encourage them to message Homework Help, if they need support.

WEEK 9 45m Chemistry Lessons - Lesson 123

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.15.3-15.4

- ∞ Video Link: Calculating Concentration (4:54)

- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 9 45m Chemistry Lessons - Lesson 124

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.15.5

- ∞ Video Link: Making Dilutions (3:41)

- Tell what you know about the objectives at the start of this section.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 9 45m Chemistry Lessons - Lesson 125

Observing Colligative Properties

Materials: High School Chemistry Lab Book and materials listed within

→ READ & NARRATE

Read the Introduction to Lab 11 and complete the introductory narration in your notebook.

- ∞ Video Link: Colligative Properties Lab (4:51)
- ∞ Image Link: Most Lab 11 materials



Term 2

WEEK 9 60m Chemistry Lessons - Lesson 126

60m Chemistry Lab - Lesson 126

Materials: High School Chemistry Lab Book and materials listed within

Observing Colligative Properties

PREP: Read Teacher Tip

→ LAB DAY

Complete Lab 11.

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 10 45m Chemistry Lessons - Lesson 127

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.15.6

∞ Video Link: Solution Stoichiometry (3:59)

- Tell what you know about the objectives at the start of this section.
- Discuss the Critical Thinking question in Sec.15-6.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 10 45m Chemistry Lessons - Lesson 128

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.15.7-15.8: If helpful, use the model kit to build the HCl and NaOH in section 15.7. Follow them during the reaction described to visualize what happens.

∞ Video Link: Stoichiometry and Normality (5:44)

- Tell what you know about the objectives at the start of these sections.

★ TEACHER TIP

The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 10 45m Chemistry Lessons - Lesson 129

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Ch.15 Review. Then choose exercises to begin.

∞ Video Link: Chapter 15 Practice (22:42).

15.3 #16, 24,
15.4 #40, 48,
15.5 #58, 61,
15.6 #66,
15.7 #72,
15.8 #88, from 9e

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 10 45m Chemistry Lessons - Lesson 130

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Continue exercises.

! STUDENT REMINDER

Don't forget to review your lesson notes!



Term 2

- Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

WEEK 10 45m Chemistry Lessons - Lesson 131

Predicting a Neutralization Reaction

Materials: High School Chemistry Lab Book and materials listed within

→ READ & NARRATE

Read the Introduction to Lab 12 and complete the introductory narration in your notebook.

∞ Video Link: Neutralization Lab (1:17)

∞ Image Link: Most Lab 12 materials

WEEK 10 60m Chemistry Lessons - Lesson 132

60m Chemistry Lab - Lesson 132

Materials: High School Chemistry Lab Book and materials listed within

Predicting a Neutralization Reaction

PREP: Read Teacher Tip

→ LAB DAY

Complete the Procedure for Lab 12.

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 11 45m Chemistry Lessons - Lesson 133

Materials: High School Chemistry Lab Book

Introductory Chemistry

→ REPORT WRITING & REVIEW

Spend this week completing a lab report for Lab 11, then review what you have learned in this course. Review and edit the lab report before turning it in. While editing, notice your tense. Most of your report should be written in the past tense because it describes and explains something that already happened. The Introduction is often in the present tense, however. Sometimes, parts of the Discussion will be in the present or future tense. Consider what you are communicating to the reader with your tense.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 11 45m Chemistry Lessons - Lesson 134

Materials: High School Chemistry Lab Book

Introductory Chemistry

PREP: Read Teacher Tip

→ REPORT WRITING & REVIEW

Spend this week completing a lab report for Lab 11, then review what you have learned so far in this course.

★ TEACHER TIP

When student(s) take exams next week, evaluate what they recall, but more importantly, whether they can explain their thinking and demonstrate a habit of problem-solving. If no relational growth is observed, consider why: do they need more practice, more support, less business, etc? Reach out through Contact Us, if you need help.



Term 2

WEEK 11 45m Chemistry Lessons - Lesson 135

Materials: High School Chemistry Lab Book

Introductory Chemistry

→ REPORT WRITING & REVIEW

Spend this week completing a lab report for Lab 11, then review what you have learned so far in this course.

WEEK 11 45m Chemistry Lessons - Lesson 136

Materials: High School Chemistry Lab Book

Introductory Chemistry

→ REPORT WRITING & REVIEW

Spend this week completing a lab report for Lab 11, then review what you have learned so far in this course.

WEEK 11 45m Chemistry Lessons - Lesson 137

Materials: High School Chemistry Lab Book

Introductory Chemistry

→ REPORT WRITING & REVIEW

Spend this week completing a lab report for Lab 11, then review what you have learned so far in this course.

WEEK 11 60m Chemistry Lessons - Lesson 138

60m Chemistry Lab - Lesson 138

Materials: High School Chemistry Lab Book

Introductory Chemistry

→ REPORT WRITING & REVIEW

Spend this week completing a lab report for Lab 11, then review what you have learned so far in this course.

WEEK 12 45m Chemistry Lessons - Lesson 139

Exam Week

→ EXAMS

Answer question(s) related to course. Complete about three questions per day or as directed by your teacher. You may use the reference material inside the covers of your book as needed.

WEEK 12 45m Chemistry Lessons - Lesson 140

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Chemistry Lessons - Lesson 141

Exam Week

→ EXAMS

Answer question(s) related to course.



Term 2

WEEK 12 45m Chemistry Lessons - Lesson 142

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Chemistry Lessons - Lesson 143

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 60m Chemistry Lessons - Lesson 144

60m Chemistry Lab - Lesson 144

Exam Week

→ EXAMS

Answer question(s) related to course.



Term 3

WEEK 1 45m Chemistry Lessons - Lesson 145

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ REMINDER

Use the video lessons in whatever way is most helpful to you. You can turn on or off closed captions, speed them up or slow them down, use the pause button, etc. Some students prefer to watch the video lesson first and refer to the book as needed, some prefer to read the lesson first and watch the video as a summary or visual recap, and some don't like video lessons and don't use them at all. If you would like additional resources, you can find alternative videos in your Quick Links, as well as the Homework Help link to message Mrs. Merritt.

→ READ, NARRATE, & DISCUSS

Ch.16.1-16.2

∞ Video Link: Acids and Bases (5:54)

- Tell what you know about the objectives at the start of these sections.
- Build the acids at the end of section 16.2 and identify the acidic protons.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

★ TEACHER TIP

The linked videos and the objectives at the start of each section are a great way for teachers to keep up with what students are learning, even when they can't read the book alongside students! If you prefer a slower, more conceptual approach without the problem-solving and lab tie-in, there are some alternatives in the Quick Links.

WEEK 1 45m Chemistry Lessons - Lesson 146

Materials: Introductory Chemistry

→ NOTE

Logarithms are introduced in Sec.16.4. If you have not learned logarithms yet in math, that is okay. Only a conceptual understanding is needed at this time, which this text will teach you. As you read 16.4, notice that logarithms are the inverse operation of exponents (just as subtraction is the inverse operation of addition and division is the inverse operation of multiplication). With this idea of inverse operations, try to understand how the pH scale is related to the concentration of hydrogen cations.

→ READ, NARRATE, & DISCUSS

Ch.16.3-16.4

∞ Video Link: The pH Scale (7:07)

- Tell what you know about the objectives at the start of these sections.
- Discuss the Critical Thinking question in Sec.16-4.

WEEK 1 45m Chemistry Lessons - Lesson 147

Materials: Introductory Chemistry

→ NOTE

Learners not yet ready to manipulate logarithms may skip Sec.16.5 and associated practice problems. If you want to go over the basics of logarithms as they relate to pH, consider watching the Math Break video:

∞ Video Link: Math Break: Logarithms (6:54)

→ READ, NARRATE, & DISCUSS

Ch.16.5-16.6: Visualize buffering by building at least 3 acetic acid molecules with your kit (the structure is at the end of section 16.2) and imagine putting these into water with the lesson. Since it is a weak acid, separate only 1 acidic proton. What happens when you add HCl, a strong acid, to this mixture? What happens when you add NaOH, a strong base?

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!



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∞ Video Link: Buffered Solutions (3:25)

- Tell what you know about the objectives at the start of these sections.

WEEK 1 45m Chemistry Lessons - Lesson 148

📄 Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Ch.16 Review. Then choose exercises to begin, skipping #41-54 and 57-58 if not ready to manipulate logarithms. You will have more time next week to complete these.

∞ Video Link: Chapter 16 Practice (34:42).

Act. #3,

16.1 #10, 14, 16,

16.2 #20, 25,

16.3 #32, 36,

16.4 #42, 44, 52, from 9e

! STUDENT REMINDER
Don't forget to review your lesson notes!

WEEK 1 45m Chemistry Lessons - Lesson 149

Analyzing the Content of Lemon Juice

📄 Materials: High School Chemistry Lab Book and materials listed within

→ READ & NARRATE

Read the Introduction to Lab 13 and complete the introductory narration in your notebook.

∞ Video Link: Lemon Juice Lab (2:59)

∞ Image Link: Most Lab 13 materials

WEEK 1 60m Chemistry Lessons - Lesson 150

60m Chemistry Lab - Lesson 150

📄 Materials: Website link, container to collect water sample, nitrate and pH test strips, optional: dip net

PREP: Read Teacher Tip

→ FIELD WORK

Complete the first collection and analysis for your citizen science project: collect a water sample from your chosen location, dip each test strip into the sample in turn, compare the test strip to the color key on the package to record your data, then make any observations regarding the local geography, plant and animal life, etc.

∞ Website Link: Clean Water Hub

∞ Website Link: Macroinvertebrate Gallery

∞ Website Link: Macroinvertebrate ID Guide

★ TEACHER TIP
Ask students to tell about their citizen science project. What data did they collect? What is the purpose?

WEEK 2 45m Chemistry Lessons - Lesson 151

📄 Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ REVIEW AND PRACTICE

Continue exercises.

- Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

• PLAN WEEKLY
📄 nature walk - record observations
📄 science free read

★ TEACHER TIP
Use the objectives at the start of assigned sections to engage in discussion and note progress, even when students are working independently.



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WEEK 2 45m Chemistry Lessons - Lesson 152

Materials: Current Events link

Introductory Chemistry

→ READ, NARRATE, & DISCUSS

What interests you? Find out what chemistry can teach you about that interest by searching for it or browse to discover something new at one of the current event resources below:

- ∞ Website Link: Science News for Students
- ∞ Website Link: Chemistry at Science Daily
- ∞ Website Link: Chemical & Engineering News

→ READ, NARRATE, & DISCUSS

Ch.17.1-17.2

- ∞ Video Link: Collision Theory (3:33)

- Tell what you know about the objectives at the start of these sections.
- Choose at least one Critical Thinking question in Sec.17-1 or 17-2.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

WEEK 2 45m Chemistry Lessons - Lesson 153

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.17.3-17.4

- ∞ Video Link: Chemical Equilibrium (2:52)

- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 2 45m Chemistry Lessons - Lesson 154

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.17.5-17.6

- ∞ Video Link: The Equilibrium Constant (5:05)

- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 2 45m Chemistry Lessons - Lesson 155

Analyzing the Content of Lemon Juice

Materials: High School Chemistry Lab Book and materials listed within

→ LAB DAY

Begin Lab 13.

WEEK 2 60m Chemistry Lessons - Lesson 156

60m Chemistry Lab - Lesson 156

Materials: High School Chemistry Lab Book and materials listed within

Analyzing the Content of Lemon Juice

PREP: Read Teacher Tip

→ LAB DAY

Complete Lab 13.

★ TEACHER TIP

Ask students to explain their lab to you!



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WEEK 3 45m Chemistry Lessons - Lesson 157

Materials: High School Chemistry Lab Book

→ REPORT WRITING

Write a lab report for Lab 13. Review and edit before turning it in. Use complete sentences, good grammar, third-person voice, correct tense, significant figures, and scientific notation, as appropriate. Remember to read the report as if you were another scientist unfamiliar with the experiment. Does the report make sense? Would you be able to repeat the experiment using the lab report? If not, edit the report to make it so.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 3 45m Chemistry Lessons - Lesson 158

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.17.7

∞ Video Link: Le Châtelier's Principle (4:31)

- Tell what you know about the objectives at the start of this section.

★ TEACHER TIP

The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 3 45m Chemistry Lessons - Lesson 159

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.17.8-17.9

∞ Video Link: Using K (4:07)

- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 3 45m Chemistry Lessons - Lesson 160

Materials: Introductory Chemistry

→ REVIEW AND PRACTICE

Ch.17 Review. Then choose exercises to begin. You will have more time next week to complete these.

∞ Video Link: Chapter 17 Practice (25:05).

17.5 #18, 22,

17.6 #26,

17.7 #35, 40,

17.8 #46, 50,

17.9 #60, 64, 68, from 9e

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 3 45m Chemistry Lessons - Lesson 161

Analyzing Reaction Kinetics

Materials: High School Chemistry Lab Book and materials listed within

→ READ & NARRATE

Read the Introduction to Lab 14 and complete the introductory narration in your



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	notebook. ∞ Video Link: Kinetics (1:05) ∞ Image Link: Most Lab 14 materials	
WEEK 3	60m Chemistry Lessons - Lesson 162 <i>60m Chemistry Lab - Lesson 162</i> Materials: High School Chemistry Lab Book and materials listed within Analyzing Reaction Kinetics PREP: Read Teacher Tip → LAB DAY Complete Lab 14.	★ TEACHER TIP Ask students to explain their lab to you!
WEEK 4	45m Chemistry Lessons - Lesson 163 Materials: Introductory Chemistry → REVIEW AND PRACTICE Continue exercises. Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?	! STUDENT REMINDER Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion. • PLAN WEEKLY - nature walk - record observations - science free read
WEEK 4	45m Chemistry Lessons - Lesson 164 Materials: Introductory Chemistry PREP: Read Teacher Tip → READ, NARRATE, & DISCUSS Ch.18.1-18.2 ∞ Video Link: Reduction and Oxidation (5:03) - Tell what you know about the objectives at the start of these sections. - Discuss the Critical Thinking question in Sec.18-2.	★ TEACHER TIP Don't forget to check that students are using the objectives at the start of assigned sections to assess their learning. Encourage them to message Homework Help, if they need support.
WEEK 4	45m Chemistry Lessons - Lesson 165 Materials: Introductory Chemistry → READ, NARRATE, & DISCUSS Ch.18.3 ∞ Video Link: Redox between Nonmetals (3:57) - Tell what you know about the objectives at the start of this section.	
WEEK 4	45m Chemistry Lessons - Lesson 166 Materials: Introductory Chemistry → READ, NARRATE, & DISCUSS Ch.18.4 ∞ Video Link: Balancing Redox Reactions (4:46) - Tell what you know about the objectives at the start of this section.	! STUDENT REMINDER Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!



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WEEK 4 45m Chemistry Lessons - Lesson 167

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.18.5-18.7

∞ Video Link: Electrochemistry (7:36)

- Discuss the Critical Thinking question in Sec.18-7.
- Tell what you know about the objectives at the start of these sections.

WEEK 4 60m Chemistry Lessons - Lesson 168

60m Chemistry Lab - Lesson 168

Materials: Introductory Chemistry

→ CATCH UP & REVIEW

Complete any unfinished work, then do a cumulative review of all you have learned. As you look back over your notes and narrate objectives from the chapters, reflect on whether the ideas make sense or seem confusing. If anything is confusing, read back over the associated section(s), try some exercises, and/or use Homework Help.

WEEK 5 45m Chemistry Lessons - Lesson 169

Materials: Introductory Chemistry

→ NOTE

This section is more about electrolysis as the reverse reaction to that in the galvanic cell, which was already touched on in the last reading and in our video. Read it if interested, but you do not need to spend a lot of time with it.

→ READ, NARRATE, & DISCUSS

Ch.18.8.

→ REVIEW AND PRACTICE

Ch.18 Review. Then choose exercises to begin.

∞ Video Link: Chapter 18 Practice (32:42).

18.1 #6,
18.2 #14, 18, 22,
18.3 #34, 36,
18.4 #42, 46,
18.5 #54, from 9e

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 5 45m Chemistry Lessons - Lesson 170

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ REVIEW AND PRACTICE

Continue exercises.

- Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

★ TEACHER TIP

Learners encounter a lot more technical content at this disciplinary stage. If they do not remember or fully understand every detail, that is okay. Can they talk about it and explain their thinking? Do they know how to go back and find what they need if they need it? Are they able to lean into the uncertainty with curiosity and a habit of problem-solving? If you are



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		unsure about their progress, reach out through Contact Us.
WEEK 5 45m Chemistry Lessons - Lesson 171 Materials: Introductory Chemistry → READ, NARRATE, & DISCUSS Ch.19.1-19.2 ∞ Video Link: Radioactivity (6:32) • Tell what you know about the objectives at the start of these sections. • Discuss the Critical Thinking question in Sec.19-1.		! STUDENT REMINDER Don't forget to review your lesson notes!
WEEK 5 45m Chemistry Lessons - Lesson 172 Materials: Introductory Chemistry → READ, NARRATE, & DISCUSS Ch.19.3-19.4 ∞ Video Link: Half-Life (3:16) • Tell what you know about the objectives at the start of these sections. ∞ Video Link: How does radiocarbon dating work? (2:10)		! STUDENT REMINDER Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!
WEEK 5 45m Chemistry Lessons - Lesson 173 <i>Producing Hydrogen and Oxygen from Water</i> Materials: High School Chemistry Lab Book and materials listed within → READ & NARRATE Read the Introduction to Lab 15 and complete the introductory narration in your notebook. ∞ Video Link: Electrolysis Lab (3:36) ∞ Image Link: Most Lab 15 materials		
WEEK 5 60m Chemistry Lessons - Lesson 174 <i>60m Chemistry Lab - Lesson 174</i> Materials: High School Chemistry Lab Book and materials listed within Producing Hydrogen and Oxygen from Water PREP: Read Teacher Tip → LAB DAY Complete Lab 15.		★ TEACHER TIP Ask students to explain their lab to you!
WEEK 6 45m Chemistry Lessons - Lesson 175 Materials: Website link Introductory Chemistry → READ, NARRATE, & DISCUSS What interests you? Find out what chemistry can teach you about that interest by searching for it or browse to discover something new at one of the current event resources below: ∞ Website Link: Science News for Students		! STUDENT REMINDER Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion. • PLAN WEEKLY nature walk - record



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∞ Website Link: Chemistry at Science Daily ∞ Website Link: Chemical & Engineering News → READ, NARRATE, & DISCUSS Ch.19.5	observations ☐ science free read
WEEK 6 ☐ 45m Chemistry Lessons - Lesson 176 ☐ Materials: Introductory Chemistry PREP: Read Teacher Tip → READ, NARRATE, & DISCUSS Ch.19.6-19.10 ∞ Video Link: Nuclear Energy (4:54) • Tell what you know about the objectives at the start of these sections. • Discuss the Critical Thinking question in Sec.19-8.	★ TEACHER TIP The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!
WEEK 6 ☐ 45m Chemistry Lessons - Lesson 177 ☐ Materials: Introductory Chemistry → REVIEW AND PRACTICE Ch.19 Review. Then choose exercises to begin. ∞ Video Link: Chapter 19 Practice (11:14). Act. #2, 19.1 #22, 28, 19.3 #40, 42, from 9e	! STUDENT REMINDER Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!
WEEK 6 ☐ 45m Chemistry Lessons - Lesson 178 ☐ Materials: Introductory Chemistry → REVIEW AND PRACTICE Continue exercises. • Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?	
WEEK 6 ☐ 45m Chemistry Lessons - Lesson 179 ☐ Materials: Introductory Chemistry → CATCH UP & REVIEW Complete any unfinished work, then do a cumulative review of all you have learned. As you look back over your notes and narrate objectives from the chapters, reflect on whether the ideas make sense or seem confusing. If anything is confusing, read back over the associated section(s), try some exercises, and/or use Homework Help.	
WEEK 6 ☐ 60m Chemistry Lessons - Lesson 180 <i>60m Chemistry Lab - Lesson 180</i> ☐ Materials: Website link, container to collect water sample, nitrate and pH test strips, optional: dip net PREP: Read Teacher Tip	★ TEACHER TIP Ask students to tell about their citizen science project. Was their data different today or the same as the first collection? Did they notice anything different at the site?



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→ FIELD WORK

Complete the second collection and analysis for your citizen science project: collect a water sample from your chosen location, dip each test strip into the sample in turn, compare the test strip to the color key on the package to record your data, then make any observations regarding the local geography, plant and animal life, etc.

- ∞ Website Link: Clean Water Hub
- ∞ Website Link: Macroinvertebrate Gallery
- ∞ Website Link: Macroinvertebrate ID Guide

WEEK 7 45m Chemistry Lessons - Lesson 181

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.20.1-20.2: Use your model kit!

∞ Video Link: Alkanes (4:51)

- Tell what you know about the objectives at the start of these sections. Use your model kit as much as is helpful throughout this chapter!

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 7 45m Chemistry Lessons - Lesson 182

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.20.3-20.4: Use your model kit!

∞ Video Link: Branched Alkanes (9:55)

- Tell what you know about the objectives at the start of these sections.

★ TEACHER TIP

The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 7 45m Chemistry Lessons - Lesson 183

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.20.5-20.7

∞ Video Link: Alkenes and Alkynes (7:29)

- Tell what you know about the objectives at the start of these sections.
- Discuss the Critical Thinking question in Sec.20-5.

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 7 45m Chemistry Lessons - Lesson 184

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.20.8-20.9

∞ Video Link: Rings (8:22)

- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!



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WEEK 7 45m Chemistry Lessons - Lesson 185

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.20.10-20.12

∞ Video Link: Functional Groups Part 1 (4:52)

- Tell what you know about the objectives at the start of these sections.

WEEK 7 60m Chemistry Lessons - Lesson 186

60m Chemistry Lab - Lesson 186

Materials: High School Chemistry Lab Book and materials listed within

Hydrolyzing Esters

PREP: Read Teacher Tip

→ READ & NARRATE

Read the Introduction to Lab 16 and complete the introductory narration in your notebook.

∞ Video Link: Soap Lab (5:30)

∞ Image Link: Most Lab 16 materials

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 8 45m Chemistry Lessons - Lesson 187

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.20.13-20.14

∞ Video Link: Functional Groups Part 2 (4:06)

- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 8 45m Chemistry Lessons - Lesson 188

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.20.15-20.16: Use your model kit!

∞ Video Link: Functional Groups Part 3 (8:00)

- Tell what you know about the objectives at the start of these sections.

★ TEACHER TIP

The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 8 45m Chemistry Lessons - Lesson 189

Materials: Introductory Chemistry

→ NOTE

Review and practice for the last two chapters of the book are largely about applying the rules and definitions from the chapter. Look back at the rules and definitions in the text as much as you need to practice. Use Homework Help if you need help.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!



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→ REVIEW AND PRACTICE Ch.20 Review. Then choose exercises to begin.	
WEEK 8 45m Chemistry Lessons - Lesson 190 Materials: Introductory Chemistry → REVIEW AND PRACTICE Continue exercises. Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?	! STUDENT REMINDER Don't forget to review your lesson notes!
WEEK 8 45m Chemistry Lessons - Lesson 191 <i>Hydrolyzing Esters</i> Materials: High School Chemistry Lab Book and materials listed within → LAB DAY Begin Lab 16. Note that the mixing phase of this experiment can take 1-2 hours, depending on temperature and mixing efficiency.	
WEEK 8 60m Chemistry Lessons - Lesson 192 <i>60m Chemistry Lab - Lesson 192</i> Materials: High School Chemistry Lab Book and materials listed within Hydrolyzing Esters PREP: Read Teacher Tip → LAB DAY Continue Lab 16.	★ TEACHER TIP Ask students to explain their lab to you!
WEEK 9 45m Chemistry Lessons - Lesson 193 <i>Hydrolyzing Esters</i> Materials: High School Chemistry Lab Book and materials listed within → LAB DAY Complete Lab 16.	! STUDENT REMINDER Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion. • PLAN WEEKLY - nature walk - record observations - science free read
WEEK 9 45m Chemistry Lessons - Lesson 194 Materials: Introductory Chemistry → REVIEW AND PRACTICE Continue exercises.	
WEEK 9 45m Chemistry Lessons - Lesson 195 Materials: High School Chemistry Lab Book → REPORT WRITING Write a lab report for Lab 16. Review and edit before turning it in. Use complete	



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sentences, good grammar, third-person voice, correct tense, and significant figures and scientific notation, as appropriate. Remember to read as if you were another scientist who is unfamiliar with the experiment. Does the report make sense? Would you be able to repeat the experiment using the lab report? If not, edit the report to make it so.

WEEK 9 45m Chemistry Lessons - Lesson 196

Materials: Introductory Chemistry

→ CATCH UP & REVIEW

Complete any unfinished work, then do a cumulative review of all you have learned. As you look back over your notes and narrate objectives from the chapters, reflect on whether the ideas make sense or seem confusing. If anything is confusing, read back over the associated section(s), try some exercises, and/or use Homework Help.

WEEK 9 45m Chemistry Lessons - Lesson 197

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.21.1-21.4: Build the amino acid shown in section 21.2 (you can use any single atom that you want to represent the R group). Identify the functional groups on this amino acid. Then you can play with building some of the R groups shown in Figure 21.2. Very large structures are needed to demonstrate secondary and tertiary structure. How could you visualize these?

∞ Video Link: Protein Structure (6:09)

- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 9 60m Chemistry Lessons - Lesson 198

60m Chemistry Lab - Lesson 198

Materials: High School Chemistry Lab Book and materials listed within

Polymerizing with Borax

PREP: Read Teacher Tip

→ LAB DAY

Complete Lab 17.

∞ Video Link: Silly Putty Lab (4:37)

∞ Image Link: Most Lab 17 materials

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 10 45m Chemistry Lessons - Lesson 199

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.21.5-21.6

∞ Video Link: Protein Function (3:46)

- Tell what you know about the objectives at the start of these sections.
- The enzyme-substrate mechanism described in your book is very rudimentary. To learn about the more commonly accepted, dynamic model, watch the linked Induced Fit video.
- ∞ Video Link: Induced Fit (6:40)

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



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WEEK 10 45m Chemistry Lessons - Lesson 200

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Ch.21.7: Use your model kit to build a carbohydrate. Compare and contrast with the amino acid.

∞ Video Link: Carbohydrates (5:01)

- Tell what you know about the objectives at the start of this section.

★ TEACHER TIP

The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 10 45m Chemistry Lessons - Lesson 201

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.21.8

∞ Video Link: Nucleic Acids (4:41)

- Tell what you know about the objectives at the start of this section.
- Can you use your model kit to build a nucleic acid? What features make something a nucleic acid? How else could you model this large molecule?

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 10 45m Chemistry Lessons - Lesson 202

Chelating with Sodium Alginate

Materials: High School Chemistry Lab Book and materials listed within

→ LAB DAY

Begin Lab 18.

∞ Video Link: Dessert Caviar Lab (5:30)

∞ Image Link: Most Lab 18 materials

WEEK 10 45m Chemistry Lessons - Lesson 203

Materials: Introductory Chemistry

→ READ, NARRATE, & DISCUSS

Ch.21.9

∞ Video Link: Lipids (6:15)

- Tell what you know about the objectives at the start of this section.
- Tell how structure produces the common property shared by all lipids.

→ REVIEW AND PRACTICE

Review and practice for the last two chapters of the book are largely about applying the rules and definitions from the chapter. Look back at the rules and definitions in the text as much as you need to practice.

Ch.21 Review. Then choose exercises to begin.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 10 60m Chemistry Lessons - Lesson 204

60m Chemistry Lab - Lesson 204

Materials: High School Chemistry Lab Book and materials listed within

Chelating with Sodium Alginate

PREP: Read Teacher Tip

★ TEACHER TIP

Ask students to explain their lab to you!



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→ LAB DAY

Complete Lab 18.

WEEK 11 45m Chemistry Lessons - Lesson 205

Materials: Introductory Chemistry

Website link, container to collect water sample, nitrate and pH test strips, optional: dip net

→ WRAP UP & REVIEW

Spend this week completing Lab 19, Ch.21 exercises, your third Citizen Science collection (and data upload), and reviewing what you have learned in this course. Remember that your data collection is as follows: collect a water sample from your chosen location, dip each test strip into the sample in turn, compare the test strip to the color key on the package to record your data, then make any observations regarding the local geography, plant and animal life, etc.

As part of your citizen science project, create a poster or graphic summarizing the water health at your test site. Remember that you want to think about how to communicate your results effectively! You can do this on paper or in a digital program, like Canva, turning it in as part of your exam.

- ∞ Website Link: Clean Water Hub
- ∞ Website Link: Macroinvertebrate Gallery
- ∞ Website Link: Macroinvertebrate ID Guide
- ∞ Video Link: Meringue Lab (4:59)
- ∞ Image Link: Most Lab 19 materials

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 11 45m Chemistry Lessons - Lesson 206

Materials: Introductory Chemistry

Citizen Science Project

PREP: Read Teacher Tip

→ WRAP UP & REVIEW

Spend this week completing Lab 19, Ch.21 exercises, your third Citizen Science collection (and data upload), and reviewing what you have learned in this course.

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 11 45m Chemistry Lessons - Lesson 207

Materials: Introductory Chemistry

Citizen Science Project

PREP: Read Teacher Tip

→ WRAP UP & REVIEW

Spend this week completing Lab 19, Ch.21 exercises, your third Citizen Science collection (and data upload), and reviewing what you have learned in this course.

★ TEACHER TIP

Ask students to tell about their citizen science project. Was the data different today or the same? What does that mean? Why do they think that? Did they notice anything different at the site this week?

WEEK 11 45m Chemistry Lessons - Lesson 208

Materials: Introductory Chemistry

Citizen Science Project

PREP: Read Teacher Tip

→ WRAP UP & REVIEW

Spend this week completing Lab 19, Ch.21 exercises, your third Citizen Science collection (and data upload), and reviewing what you have learned in this course.

★ TEACHER TIP

When student(s) take exams next week, evaluate what they recall, but more importantly, whether they can explain their thinking and demonstrate a habit of problem-solving. If no relational growth is observed, consider why: do they need



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more practice, more support, less business, etc? Reach out through Contact Us, if you need help.

WEEK 11 45m Chemistry Lessons - Lesson 209

Materials: Introductory Chemistry

Citizen Science Project

→ WRAP UP & REVIEW

Spend this week completing Lab 19, Ch.21 exercises, your third Citizen Science collection (and data upload), and reviewing what you have learned in this course.

WEEK 11 60m Chemistry Lessons - Lesson 210

60m Chemistry Lab - Lesson 210

Materials: Introductory Chemistry

Citizen Science Project

→ WRAP UP & REVIEW

Spend this week completing Lab 19, Ch.21 exercises, your third Citizen Science collection (and data upload), and reviewing what you have learned in this course.

WEEK 12 45m Chemistry Lessons - Lesson 211

Exam Week

→ EXAMS

Answer question(s) related to course. Complete about three questions per day or as directed by your teacher. You may use the reference material inside the covers of your book as needed.

WEEK 12 45m Chemistry Lessons - Lesson 212

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Chemistry Lessons - Lesson 213

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Chemistry Lessons - Lesson 214

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Chemistry Lessons - Lesson 215

Exam Week

→ EXAMS

Answer question(s) related to course.



Term 3

WEEK 12 60m Chemistry Lessons - Lesson 216

60m Chemistry Lab - Lesson 216

Exam Week

→ EXAMS

Answer question(s) related to course.

Science: Chemistry

Examination

Term 1

Chemistry Lessons

- Day 1: Choose 2 of the following from Introductory Chemistry:
 - #8, #12, #24, #26 from Ch.4-5 Cumulative Rev
- Day 2: Choose 2 of the following from Introductory Chemistry (substitute previous day's questions as appropriate based on material covered):
 - #12, #16, #10, #18 from Ch.6-7 Cumulative Rev
- Day 3: Choose 2 of the following from Introductory Chemistry (substitute previous days' questions as appropriate based on material covered):
 - #2, #14, #18, #20 from Ch.8-9 Cumulative Rev
- Day 4: Explain one of your laboratory exercises from this term and what you learned from it. Describe any questions this made you think of.
- Day 5: Explain an example of chemistry in the world around you that you noticed this term. This might be from your project work, current events, nature walks, or your free read. Describe any questions this made you think of.

Term 2

Chemistry Lessons

- Day 1: Choose 2 of the following from Introductory Chemistry (substitute previous term's questions as appropriate based on material covered):
 - #2, #14, #20, #24 from Ch.10-12 Cumulative Rev
- Day 2: Choose 2 of the following from Introductory Chemistry (substitute previous questions as appropriate based on material covered):
 - #28, #34, #48 from Ch.10-12 Cumulative Rev, #4 from Ch.13-15 Cumulative Rev
- Day 3: Choose 2 of the following from Introductory Chemistry (substitute previous questions as appropriate based on material covered):
 - #14, #22, #32, #36 from Ch.13-15 Cumulative Rev
- Day 4: Explain one of your laboratory exercises from this term and what you learned from it. Describe any questions this made you think of.
- Day 5: Explain an example of chemistry in the world around you that you noticed this term. This might be from your project work, current events, nature walks, or your free read. Describe any questions this made you think of.

Term 3

Chemistry Lessons

- Day 1: Choose 2 of the following from Introductory Chemistry (substitute previous term's questions as appropriate based on material covered):
 - #2, #10, #14, #22 from Ch.16-17 Cumulative Rev
- Day 2: Choose 2 of the following from Introductory Chemistry (substitute previous questions as appropriate based on material covered):
 - #84, #94 from Ch.18, #92, #96 from Ch.19
- Day 3: Choose 2 of the following from Introductory Chemistry (substitute previous questions as appropriate based on material covered):
 - #118, #134 from Ch.20, #14, #42 from Ch.21
- Day 4: Explain one of your laboratory exercises from this term and what you learned from it. Describe any questions this made you think of.
- Day 5: Submit your citizen science poster or graphic, supplemented with any stories or explanations that are helpful. Convey any questions this project made you think of.